



# UNITED STATES PATENT AND TRADEMARK OFFICE

UNITED STATES DEPARTMENT OF COMMERCE  
United States Patent and Trademark Office

Address: COMMISSIONER FOR PATENTS  
P.O. Box 1450  
Alexandria, Virginia 22313-1450  
www.uspto.gov

| APPLICATION NO. | FILING DATE | FIRST NAMED INVENTOR    | ATTORNEY DOCKET NO. | CONFIRMATION NO. |
|-----------------|-------------|-------------------------|---------------------|------------------|
| 18/776,642      | 07/18/2024  | Richard A. Giambruno II | 1009-0002           | 7765             |

185208 7590 04/21/2026

Kiklis and Clark, PLLC  
1800 Diagonal Road  
Suite 600  
Alexandria, VA 22314

EXAMINER

SHUI, MING

ART UNIT

PAPER NUMBER

2663

NOTIFICATION DATE

DELIVERY MODE

04/21/2026

ELECTRONIC

**Please find below and/or attached an Office communication concerning this application or proceeding.**

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

docket@kiklisclark.com

info@kiklisclark.com

kimani@kiklisclark.com

## Office Action Summary

**Application No.**

18/776,642

**Applicant(s)**

Giambruno, Richard A.

**Examiner**

Ming Shui

**Art Unit**

2663

**AIA (First Inventor to File) Status Yes**

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

**Period for Reply**

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

**Status**

1) ☒ Responsive to communication(s) filed on 7/18/24.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on \_\_\_\_.

2a) ☐ This action is **FINAL**.

2b) ☒ This action is non-final.

3) ☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on \_\_\_\_; the restriction requirement and election have been incorporated into this action.

4) ☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

**Disposition of Claims\***

5) ☒ Claim(s) 1-20 is/are pending in the application.

5a) Of the above claim(s) \_\_\_\_ is/are withdrawn from consideration.

6) ☐ Claim(s) \_\_\_\_ is/are allowed.

7) ☒ Claim(s) 1-20 is/are rejected.

8) ☐ Claim(s) \_\_\_\_ is/are objected to.

9) ☐ Claim(s) \_\_\_\_ are subject to restriction and/or election requirement

\* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see [http://www.uspto.gov/patents/init\\_events/pph/index.jsp](http://www.uspto.gov/patents/init_events/pph/index.jsp) or send an inquiry to [PPHfeedback@uspto.gov](mailto:PPHfeedback@uspto.gov).

**Application Papers**

10) ☐ The specification is objected to by the Examiner.

11) ☒ The drawing(s) filed on 7/18/24 is/are: a) ☒ accepted or b) ☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

**Priority under 35 U.S.C. § 119**

12) ☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

**Certified copies:**

a) ☐ All      b) ☐ Some\*\*      c) ☐ None of the:

1. ☐ Certified copies of the priority documents have been received.

2. ☐ Certified copies of the priority documents have been received in Application No. \_\_\_\_.

3. ☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

\*\* See the attached detailed Office action for a list of the certified copies not received.

**Attachment(s)**

1) ☒ Notice of References Cited (PTO-892)

3) ☐ Interview Summary (PTO-413)

Paper No(s)/Mail Date \_\_\_\_.

2) ☐ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b)

4) ☐ Other: \_\_\_\_.

Paper No(s)/Mail Date \_\_\_\_.

***Notice of Pre-AIA or AIA Status***

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

**DETAILED ACTION**

***Claim Rejections - 35 USC § 103***

The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102 of this title, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

**Claims 1-2, 4-5, 10-12, 14-15, and 20 are rejected under 35 USC 103 as being unpatentable over US 2019/0039613 Lee in view of How Autonomous Vehicles Work: the Self driving stack, Mobileye, May 31, 2023. (hereafter Mobileye)**

Art Unit: 2663

1. A method in a data processing system for detecting a law enforcement vehicle, the method comprising,

using a processor of the data processing system: receiving a digital image, at the processor, from a camera in electronic communication with the processor, wherein the camera is disposed within a user vehicle; (Lee ¶21 sensor is a camera and is processed by a processor, thus there is a digital image and the apparatus is in the car see e.g. Fig 4A)

detecting the law enforcement vehicle within the digital image by the processor; and (Lee ¶21 uses the sensor to detect emergency vehicles, ¶21 discloses that emergency vehicles include police cars)

transmitting, by the processor, a notification instruction to a notification output device disposed within the user vehicle indicating the law enforcement vehicle was detected. (Lee ¶24 outputs information about the detection of an emergency vehicle)

Lee does not disclose

using a machine learning model, the machine learning model trained by training data including images of law enforcement vehicles

However, Mobileye discloses using a machine learning model that uses trained object recognition, pattern, recognition, etc. to recognize particular items in the image – to recognize emergency vehicles such as law enforcement vehicles, the algorithm would have been trained with images that included law enforcement vehicles. It would have been obvious to modify the system of Lee to use machine learning because the use of machine learning is a fundamental part of creating autonomous vehicles.

Claim 11 is rejected under a similar rationale.

Art Unit: 2663

2. The method of claim 1, wherein the processor, the camera, and the notification output device are disposed within a housing. (Lee ¶69 smart phone, instrument cluster)

Claim 12 is rejected under a similar rationale.

4. The method of claim 1, further comprising training the machine learning model by the training data including images of law enforcement vehicles. (Mobileye discloses using a machine learning model that uses trained object recognition, pattern, recognition, etc. to recognize particular items in the image – to recognize emergency vehicles such as law enforcement vehicles, the algorithm would have been trained with images that included law enforcement vehicles.)

Claim 14 is rejected under a similar rationale.

5. The method of claim 1, wherein the processor, the camera, or the notification output device is integrated into the user vehicle. (Lee ¶68 included in vehicle)

Claim 15 is rejected under a similar rationale.

10. The method of claim 1, wherein the notification output device includes a mobile device, and transmitting the notification instruction to the notification output device includes transmitting a communications signal to the notification output device via a networking device. (Lee ¶69 portable device through network)

Claim 20 is rejected under a similar rationale.

**Claims 3 and 13 are rejected under 35 USC 103 as being unpatentable over Lee and Mobileye in view of US 2015/0122858, Azadi.**

Lee and Mobileye do not disclose

3. The method of claim 2, wherein the housing includes a mounting device configured to secure the housing to a dashboard of the user vehicle.

However Azadi discloses a dash mount for devices. It would have been obvious to modify the system of Lee to include a dash mount for the purposes of providing a secure location for the portable device that does not interfere with the driver's handling of the vehicle or impair the vision of the driver as taught by Azadi.

Claim 13 is rejected under a similar rationale.

**Claims 6-9 and 16-19 are rejected under 35 USC 103 as being unpatentable over Lee and Mobileye in view of US 2024/0174168 Suh.**

6. The method of claim 1, wherein the notification output device is a light device, and the notification instruction includes a computer instruction to emit light from the light device. (Suh ¶24 provides a map on a screen, which is a light device)

Art Unit: 2663

It would have been obvious to modify the system of Lee and Mobileye to include a map for the purposes of providing real time information for the car as taught by Suh.

Claim 16 is rejected under a similar rationale.

7. The method of claim 1, wherein the notification output device includes a speaker, and the notification instruction includes a computer instruction to emit a sound from the speaker. (Suh Claim 15, voice message about approaching emergency vehicle)

It would have been obvious to make the modification for the same reasons as claim 16.

Claim 17 is rejected under a similar rationale.

8. The method of claim 7, wherein the sound is a generated voice statement indicating a location of the law enforcement vehicle relative to the user vehicle. (Suh Claim 15, voice message about approaching emergency vehicle)

It would have been obvious to make the modification for the same reasons as claim 16.

Claim 18 is rejected under a similar rationale.

9. The method of claim 1, wherein the notification output device includes a display configured to display a map, and the notification instruction includes a computer instruction to display a location of the law enforcement vehicle on the map. (Suh Fig 4A illustrates the vehicle)

It would have been obvious to make the modification for the same reasons as claim 16.

Claim 19 is rejected under a similar rationale.

### ***Conclusion***

The prior art made of record and not relied upon is considered pertinent to applicant's disclosure.

Radar Detectors vs Waze: Best Speeding Ticket Countermeasure?,  
<https://www.youtube.com/watch?v=0F9EQSP5FKY>, Feb 11, 2021, disclosing a map with police symbols to notify users about speed traps.

Any inquiry concerning this communication or earlier communications from the examiner should be directed to Ming Shui whose telephone number is (303)297-4247. The examiner can normally be reached on 7-5 Pacific Time, M-Th.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Greg Morse can be reached on 571-272-3838. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.



Art Unit: 2663

Information regarding the status of an application may be obtained from the Patent Application Information Retrieval (PAIR) system. Status information for published applications may be obtained from either Private PAIR or Public PAIR. Status information for unpublished applications is available through Private PAIR only. For more information about the PAIR system, see <http://pair-direct.uspto.gov>. Should you have questions on access to the Private PAIR system, contact the Electronic Business Center (EBC) at 866-217-9197 (toll-free). If you would like assistance from a USPTO Customer Service Representative or access to the automated information system, call 800-786-9199 (IN USA OR CANADA) or 571-272-1000.

/Ming Shui/

Primary Examiner, Art Unit 2663